

Phase 1 Mini Project Report

Project Title

E-Commerce Sales Performance Analysis using Exploratory Data Analysis (EDA)

Objective

The objective of this project is to analyze e-commerce transaction data and identify key business insights related to sales performance, customer behavior, product trends, seller contribution, and payment preferences. The analysis helps management make data-driven decisions to improve overall business performance.

Dataset Description

The project uses the Brazilian E-Commerce Public Dataset (Olist) from Kaggle. The dataset contains information about customers, orders, products, payments, reviews, sellers, and product categories. Multiple datasets were merged to create a unified view for analysis and reporting.

Data Cleaning Summary

Several data preprocessing steps were performed before analysis. Missing values were identified and handled appropriately, duplicate records were checked and removed where necessary, date columns were converted into proper datetime format, and multiple datasets were merged using common keys. Data quality checks were also conducted to ensure the integrity and reliability of the final dataset.

Analysis Findings

The analysis revealed that a small number of product categories contribute a significant portion of total revenue. Major cities generated the highest sales, indicating strong market concentration in urban regions. Customer purchasing behavior showed that most customers make one-time purchases, while a smaller group of repeat customers contributes substantial business value. Credit card transactions were the most commonly used payment method. Seller analysis identified a few top-performing sellers responsible for a large share of revenue. Review score analysis highlighted categories with high customer satisfaction as well as areas requiring improvement.

Visualizations Used

- Revenue by Product Category (Bar Chart)
- Sales by City (Bar Chart)
- Payment Method Analysis (Bar Chart)
- Top Sellers Analysis (Bar Chart)
- Monthly Sales Trend (Line Chart)

- Review Score Comparison (Bar Chart)

Final Conclusion

The EDA process successfully transformed raw e-commerce data into meaningful business insights. The findings indicate opportunities to increase revenue through investment in high-performing product categories, targeted marketing in top-performing cities, and customer retention strategies for repeat buyers. Additionally, improving low-rated categories and strengthening relationships with top sellers can further enhance business performance. This project demonstrates the complete analytics workflow from data preparation to actionable business recommendations.